

AI2002 – Artificial Intelligence

Quiz –

Name	
Roll. No.	
Section	

CLO2: Apply classical AI techniques, including uninformed and informed search algorithms, minimax, and neural networks, to solve structured problems.

Q1. [6 Marks] Given the following training points for two features (x_1 , x_2) and their classes, classify the new test point $T = (5, 5)$. Show all working. Round L2 distances to 2 decimal places.

Point	x_1	x_2	Class
P1	2	6	G
P2	6	3	Y
P3	8	7	Y
P4	3	2	G

- a) Using L1 (Manhattan) distance, what is the predicted class for $k = 1$?
- b) Using L2 (Euclidean) distance, what is the predicted class for $k = 3$?

Q2. [4 Marks] A dataset contains 20,000 images. It is split into 70% training, 15% validation, and 15% testing.

- a) How many images are in the validation set?
- b) Assume a standard 5-NN implementation computes the distance from each test image to every training image before selecting the 5 nearest. If one distance computation takes 0.03 ms, how many minutes will it take to classify the entire test set?

Solutions

Q1 Solution

L1 (Manhattan) distances from T:

Point	Coordinates	Class	Distance
P1	(2, 6)	G	4.00
P2	(6, 3)	Y	3.00
P3	(8, 7)	Y	5.00
P4	(3, 2)	G	5.00

Nearest point for part (a): P2 (Y). Predicted class = Y.

L2 (Euclidean) distances from T:

Point	Coordinates	Class	Distance
P1	(2, 6)	G	3.16
P2	(6, 3)	Y	2.24
P3	(8, 7)	Y	3.61
P4	(3, 2)	G	3.61

Three nearest points for part (b): P2 (Y), P1 (G), P3 (Y). Predicted class = Y.

Q2 Solution

a) Validation images = 15% of 20,000 = 3,000.

b) Training images = 14,000; Test images = 3,000.

Total distance computations = $3,000 \times 14,000 = 42,000,000$.

Total time = $42,000,000 \times 0.03 \text{ ms} = 1,260,000 \text{ ms} = 21.00 \text{ minutes}$.

Checked final answers:

Q1(a): Y | Q1(b): Y | Q2(a): 3,000 | Q2(b): 21.00 minutes